PROOF-CARRYING SAT BARRIERS AROUND A 100-VERTEX $R(3,18)$ NEAR MISS

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Abstract. Let H be a fixed graph on 100 vertices with 827 edges, no independent set of order 18, and the unique triangle $\{97, 98, 99\}$. For a graph F on the same labelled vertex set, write $d_H(F) = |E(H) \setminus E(F)|$; pairs outside $E(H)$ are free and may be added without charge. We give a proof-carrying computation establishing that no triangle-free graph F with $\alpha(F) < 18$ satisfies $d_H(F) \leq 6$. Thus the one-sided deletion repair radius of this fixed seed is at least seven.

The proof first replays a three-branch DRAT certificate excluding deletion budget at most five. At exact budget six, triangle-freeness forces one edge of the unique seed triangle to be absent. Each of the three resulting branches is encoded as a finite CNF relaxation containing all triangle clauses, an exact-five sequential counter for the remaining seed edges, and a checked bank of independent-set hitting clauses. Independent semantic reconstruction and DRAT replay verify all three refutations. We specify the hashes, solver and checker revisions, and the boundary of machine trust. A separate exact-seven portfolio ended with all branches UNKNOWN; no existence or nonexistence claim is made at that layer. In particular, the result is local to H and does not imply $R(3, 18) \geq 101$.

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1. Introduction

Write $R(3, k)$ for the least n such that every graph on n vertices contains either a triangle or an independent set of order k. The current dynamic survey records

$$100 \leq R(3, 18) \leq 120;$$

see Radziszowski's continuously maintained table [Rad26]. A triangle-free graph on 99 vertices with independence number 17 certifies the lower endpoint. Extending such a graph by one vertex produces a natural, but logically delicate, local search around the next possible certificate.

This paper studies one labelled 100-vertex extension H. The graph has no independent 18-set but has a unique triangle. It is therefore a near miss, not a Ramsey graph. We ask a deliberately local question: how many *original* edges must be deleted before the triangle can be removed while arbitrary original nonedges are allowed to change into edges? This asymmetric edit model makes additions completely free; consequently, a lower bound in this model is stronger than a lower bound obtained by fixing the nonedges or by restricting the total symmetric-difference distance.

The principal result is a proof-carrying lower bound of seven on that local deletion radius. The argument has four independently inspectable layers.

(i) A direct graph check establishes the seed identity, its unique triangle, and the absence of an independent 18-set.
(ii) The unique triangle gives an exhaustive three-way branch split. A previous proof-carrying package excludes total deletion budget at most five, reducing the new layer to exact budget six.
(iii) Each exact-six branch is translated into a finite CNF *relaxation* of the graph problem. All structural clauses are reconstructed from the seed; the independent-set clauses are validated as universally sound.
(iv) CaDiCaL emits a DRAT refutation for each branch, and a separately pinned `drat-trim` executable verifies every refutation. Solver status strings are not used as proof evidence.

This layering is important. An unsatisfiable CNF proves the graph claim only after formula soundness and branch completeness have been established. A solver's report of unsatisfiability proves the CNF claim only after an independently checkable proof trace has been accepted. We record both implications explicitly.

The computation is also informative about what remains open. We tested the next deletion layer, exact budget seven, with one bounded solver call in each branch. Every call reached its wall-time limit before producing a separated model or a proof. We record these endpoints as UNKNOWN. They are not evidence either for or against the existence of a seven-deletion repair. Claim boundary. The theorem concerns one fixed, labelled seed and one one-sided edit metric. It neither constructs a 100-vertex $(3, 18)$-Ramsey graph nor rules out such graphs elsewhere in the search space. Thus it does not improve either endpoint of the globally tabulated interval for $R(3, 18)$. The value of the result is instead methodological and local: it gives a complete, proof-carrying obstruction around a particularly close seed, with all negative conclusions separated from incomplete search.

2. The frozen seed and the precise statement

All vertices are labelled by

$$V = \{0, 1, \ldots, 99\}.$$

The input graph H is the adjacency matrix in the artifact

`r3_18_n100_nearmiss.txt`.

Its SHA-256 identity is

`e6527601f72ebb2d3aed11adcdd2ffb0c6aab5e326d81d439c10ee468844e85e`.

Content addressing is part of the mathematical statement: replacing the matrix, even by an isomorphic relabelling, changes the frozen input unless the new identity is separately certified.

Lemma 2.1 (Seed facts)**.** *The graph H has 827 edges, satisfies $\alpha(H) < 18$, and has exactly one triangle, namely*

$$T = \{97, 98, 99\}.$$

Its first 99 vertices induce a graph G_0 with 809 edges, no triangle, and $\alpha(G_0) = 17$.

Computational proof. The matrix parser checks that the input is a symmetric 100×100 binary matrix with zero diagonal. Exhaustive enumeration of the $\binom{100}{3} = 161700$ triples finds only T. A bitset maximum-clique search on the complement returns no clique of order 18, which is equivalent to $\alpha(H) < 18$. The 99×99 principal block agrees entry-for-entry with the separately frozen matrix G_0, whose SHA-256 is

`3e9d16e29111f3a2f25ae3b992235804c2612de2f6ada5570901d17ceedfca45`.

The same verifier finds no independent 18-set in G_0 and returns an independent 17-set. It also finds no triangle in G_0. The new vertex has neighbourhood

$$N_H(99) = \{33, 34, \ldots, 48, 97, 98\},$$

and the only edge induced by this neighbourhood is $(97, 98)$, providing a second direct check of uniqueness. The negative clique routine was compared with literal subset enumeration on 8640 small random graph/target pairs; all decisions and returned witnesses agreed. Reproduction commands are given in Section 7 and Appendix B. □

The graph G_0 reproduces the already tabulated lower bound $R(3, 18) \geq 100$. The graph H, by contrast, contains a triangle and therefore does not certify $R(3, 18) \geq 101$.

Definition 2.2 (Free-addition deletion distance)**.** For a graph F on V, define

$$d_H(F) = |E(H) \setminus E(F)|.$$

Only missing input edges are charged. Every pair in $\binom{V}{2} \setminus E(H)$ may independently be added to F at zero cost. Let

$$\mathcal{F} = \{F \text{ on } V : F \text{ is triangle-free and } \alpha(F) < 18\}$$

and define

$$\rho(H) = \inf\{d_H(F) : F \in \mathcal{F}\},$$

with the convention $\inf \varnothing = \infty$.

Theorem 2.3 (Fixed-seed deletion barrier)**.** *For the content-addressed graph H above,*

$$\rho(H) \geq 7.$$

Equivalently, no $F \in \mathcal{F}$ satisfies $d_H(F) \leq 6$, even when arbitrary subsets of the 4123 *original nonedges are added.*

The convention in the definition matters: Theorem 2.3 is compatible both with $\rho(H) = 7$ and with a larger or infinite radius. We do not determine which occurs.

3. Reduction to six proof-carrying branch formulas

Let

$$f_0 = (97, 98), \qquad f_1 = (97, 99), \qquad f_2 = (98, 99)$$

be the three edges of the unique seed triangle. We first state the checked dependency at the preceding deletion layer.

Proposition 3.1 (Budget-five dependency)**.** *There is no $F \in \mathcal{F}$ with $d_H(F) \leq 5$.*

Proof-carrying computation. Every triangle-free F omits at least one f_i. In branch i, the unit $\neg x_{f_i}$ is imposed and a sequential counter permits at most four further deletions among the other 826 input edges. All triangle clauses and a finite bank of universally valid independent-18 hitting clauses are imposed. The resulting three CNFs have respectively 169382, 184967, and 177799 clauses. All three formulas were reconstructed clause-by-clause from H, and all three DRAT refutations were replayed with a freshly compiled checker from the pinned source revision. The full identities are in Table 4; the formula-soundness argument is the same as Proposition 4.1. Therefore each branch is impossible, and the three branches cover every candidate of total deletion budget at most five. □

By Proposition 3.1, a counterexample to Theorem 2.3 would have exactly six deleted input edges. Fix any one absent edge f_i of the seed triangle. The other five deletions are counted among $E(H) \setminus \{f_i\}$. Notice that the other two triangle edges are not fixed present; if either is also deleted, it simply consumes one of these five residual deletions. Thus the branches cover all candidates and may overlap.

For each $i \in \{0, 1, 2\}$, let Φ_i be the formula defined precisely in Section 4. It contains the fixed negative unit for f_i, an exact-five counter on the residual seed edges, all triangle clauses, and a branch-specific bank of independent-set hitting clauses.

Lemma 3.2 (Exact-six branch cover)**.** *If $F \in \mathcal{F}$ and $d_H(F) = 6$, then the edge assignment of F extends to a satisfying assignment of at least one of Φ_0, Φ_1, Φ_2.*

Proof. Because F is triangle-free, at least one edge f_i of T is absent. Choose such an i. The chosen unit clause holds. Exactly five members of $E(H) \setminus \{f_i\}$ are absent, so the primary edge assignment extends to the auxiliary variables of the exact-five counter. Triangle-freeness satisfies every triangle clause. Finally, $\alpha(F) < 18$ implies that each installed 18-set contains a present pair, so every installed hitting clause holds. Hence F's primary assignment extends to a model of Φ_i. □

Proposition 3.3 (Certified branch refutations)**.** *Each formula Φ_0, Φ_1, Φ_2 is unsatisfiable.*

Proof certificate. The three frozen DIMACS files have the identities and dimensions in Table 1. The corresponding DRAT traces were accepted by the pinned `drat-trim` checker, each ending with `s VERIFIED`. Before replay, independent semantic checkers reconstructed the entire triangle prefix, cardinality block, fixed unit, and every 153-literal hitting clause; they also checked the compressed and raw identities. Details are given in Section 5 and Appendix B. □

Proof of Theorem 2.3. Suppose $F \in \mathcal{F}$ and $d_H(F) \le 6$. Proposition 3.1 implies $d_H(F) = 6$. By Lemma 3.2, F's edge assignment extends to a model of at least one Φ_i. This contradicts Proposition 3.3. Therefore no such F exists and $\rho(H) \geq 7$. $\square$

4. CNF encoding and its soundness

4.1. **Primary variables and structural clauses.** For every unordered pair $e = \{u, v\} \in \binom{V}{2}$, a primary variable x_e records whether $e \in E(F)$. The 4950 primary variables are numbered in lexicographic pair order. Crucially, variables exist for all original nonedges; those variables are absent from the deletion counter and are not fixed false.

Triangle-freeness is encoded without separation: for every $a < b < c$, the formula contains

$$\neg x_{ab} \vee \neg x_{ac} \vee \neg x_{bc}.$$

There are exactly $\binom{100}{3} = 161700$ such clauses.

For every installed set $S \in \binom{V}{18}$, the formula contains the positive hitting clause

$$C_S = \bigvee_{e \in \binom{S}{2}} x_e. \tag{4.1}$$

It has $\binom{18}{2} = 153$ distinct primary literals. Every graph with independence number below 18 satisfies C_S. We install only a finite selected bank $\mathcal{B}_i$, not all $\binom{100}{18}$ clauses. Consequently, Φ_i is a relaxation of the exact Ramsey branch. This is the useful direction: unsatisfiability of the relaxation implies unsatisfiability of the exact branch.

4.2. **Exact deletion counter.** In branch i, set

$$L_i = \{\neg x_e : e \in E(H) \setminus \{f_i\}\}.$$

The list has 826 literals. A PySAT sequential-counter equality encoding implements

$$\sum_{\ell \in L_i} [\ell \text{ is true}] = 5. \tag{4.2}$$

We use the standard sequential counter of Sinz [Sin05]. The concrete generator is exposed by PySAT [IMaMS18]. It introduces 8210 auxiliary variables and 16420 clauses, ending at variable 13160. Complete primary assignments with residual deletion counts $0, 1, 4, 5, 6, 7, 825, 826$ were offered to an independent SAT backend; the auxiliary variables were extendible exactly at count five.

Thus

$$\Phi_i = \Big(\bigwedge_{a<b<c} (\neg x_{ab} \vee \neg x_{ac} \vee \neg x_{bc}) \Big) \wedge \mathrm{Card}_{=5}(L_i) \wedge \neg x_{f_i} \wedge \bigwedge_{S \in \mathcal{B}_i} C_S. \tag{4.3}$$

Proposition 4.1 (CNF soundness)**.** *For each i, every graph $F \in \mathcal{F}$ with $d_H(F) = 6$ and $f_i \notin E(F)$ induces a primary assignment that extends to a model of Φ_i.*

Table 1. Exact-six branch dimensions. All branches have 4950 primary variables, 8210 counter auxiliaries, 161700 triangle clauses, 16420 counter clauses, and one fixed unit.

branch	fixed absent edge	hitting clauses	variables	total clauses
0	$(97, 98)$	251771	13160	429892
1	$(97, 99)$	63943	13160	242064
2	$(98, 99)$	5422	13160	183543

Table 2. Audited construction of the branch-0 universal cut union.

source	source masks	new masks	cumulative
branch-0 discovery checkpoint	183133	183133	183133
branch-0 earlier bank	141149	4096	187229
branch-1 bank	63943	52515	239744
branch-2 bank	5422	3931	243675
budget-five branch 0	287	286	243961
budget-five branch 1	15872	4262	248223
budget-five branch 2	8704	3548	251771

Proof. The triangle block holds by triangle-freeness. The fixed unit holds by the branch premise. Since f_i is one of exactly six missing input edges, exactly five literals in L_i are true; completeness of the equality encoding supplies an auxiliary assignment. Each $S \in \mathcal{B}_i$ contains a present pair because $\alpha(F) < 18$, so every C_S holds. No condition was imposed on an original nonedge except through triangle and hitting clauses, hence arbitrary additions remain represented. □

4.3. **Sound reuse of the branch-0 cut union.** Branch 0 uses 251771 distinct 18-sets deduplicated from seven checked sources. The ordered mask digest is

`f10690b826b86eb03567a2ffaffb553801fae32af20cfe4337118bddf4e41afa`.

Only vertex-set masks were transferred. No learned clause, branch assumption, or solver conclusion was reused. Since every clause C_S in (4.1) follows solely from $\alpha(F) < 18$, it is valid in every branch. The source counts and cumulative deduplication totals appear in Table 2.

5. Proof production, checking, and trust

5.1. **From finite formulas to checked refutations.** The final proof-producing runs used CaDiCaL [BFF+24] at source commit

`c60730422e758ef1cebe7aeddf2dda31c996bf04`.

Table 3. Exact-six proof artifacts and replay scale. Hashes are abbreviated here and given in full in Table 5.

branch	raw DRAT bytes	proof lines	compressed bytes	replay
0	3493847713	6649939	656144450	verified
1	2137671968	5590150	433552903	verified
2	759008359	2837771	152784369	verified

An exit code 20 and the text `UNSATISFIABLE` were treated only as requests to check the emitted proofs. The final mathematical evidence is the successful replay of the DRAT traces [WHH14, Heu16] by the official `drat-trim` source at commit

`2e3b2dc0ecf938addbd779d42877b6ed69d9a985`.

The principal recorded executable has SHA-256

`31d9e8e04bc76a65f0a18ea212ac44af4cf7bc398629495cec093ced772c50f4`.

All three checks terminated with exit code zero and `s VERIFIED`.

For branch 0, a fresh replay in a temporary directory took 273.448 seconds inside the checker and reported 135152646 resolution steps in the retained core. Branch 1 reported 209897525 resolution steps in its independently checked run. These timings are descriptive, not logical premises; hashes and successful replay are the portable evidence.

5.2. **Semantic reconstruction.** The proof checker establishes unsatisfiability of the bytes in each DIMACS file. Separate semantic checkers establish that those bytes are the intended relaxations. For every branch they independently:

(1) parse the frozen seed and regenerate the lexicographic pair-variable map;
(2) compare all 161700 triangle clauses in order;
(3) regenerate the PySAT equality counter and compare its variables and clauses in order;
(4) check the branch unit and the exact clause partition;
(5) reconstruct every hitting clause as all 153 pair variables of one 18-vertex mask;
(6) verify the compressed and raw sizes and SHA-256 identities; and
(7) verify the budget-five dependency and the three-edge cover.

Branch 0 additionally reparses all seven cut sources and reproduces the deduplicated union. Branches 1 and 2 likewise reconstruct their complete stored cut banks. An earlier interrupted branch-1 trace fails gzip integrity, remains quarantined, and is excluded from every manifest and conclusion.

5.3. **Trust boundary.** The claim is proof-carrying, not formally verified in a proof assistant. Its earliest machine dependencies are:

- the bitset independent-set verifier used for the seed facts;
- the PySAT implementation of the standard sequential counter;
- the `drat-trim` proof kernel, together with its compiler and execution hardware; and
- the scripts that map matrices and masks to DIMACS.

The audit narrows this boundary through independent matrix parsing, exhaustive triangle enumeration, small-instance cross-checks of the clique routine, counter truth-table tests, complete clause reconstruction, content hashes, gzip integrity tests, and proof replay with a freshly compiled checker. A future LRAT conversion and formally verified LRAT checker could reduce the trusted kernel further, but is not required for the conventional proof-carrying SAT standard used here.

6. The unresolved exact-seven layer

The theorem implies only $\rho(H) \geq 7$. To distinguish this lower bound from an exact value, we report the first bounded experiment at deletion budget seven.

In branch i, the fixed unit $\neg x_{f_i}$ accounts for one deletion and an equality counter requires exactly six of the other 826 input edges to be absent. All 4123 original nonedges remain free. The universal bank of 251771 hitting clauses from branch 0 is valid in every branch and was preloaded. Every initial formula therefore had 4950 primary variables, 9840 counter auxiliaries, 161700 triangle clauses, 19680 counter clauses, one fixed unit, and 251771 hitting clauses, for a total of 433152 clauses.

Computational observation 6.1 (Bounded endpoints, not a theorem). One solver call was launched per branch with a parent-enforced 300-second discovery wall. Branch 0 used CaDiCaL, branch 1 Glucose 4.2, and branch 2 MapleChrono. Each remained inside its first SAT call until the wall expired. No branch produced a model that reached independent-set separation, and no branch produced a checked refutation. The endpoint of all three branches is therefore UNKNOWN.

The zero recorded separation iterations means that no SAT model returned before interruption; it does not mean the formula is unsatisfiable. The recoverable state consists of the completely reconstructed initial formula and universal cut bank. Internal CDCL state was not serialized. Exact hashes of the frozen result and checkpoint records are listed in Appendix C.

Accordingly, none of the following is established: a repair with exactly seven deletions; nonexistence of such a repair; the exact value of $\rho(H)$; or a 100-vertex $(3, 18)$-Ramsey graph. A machine-readable Boolean field whose default value is false is likewise not evidence of nonexistence unless the companion establishment flag and a checked proof support it.

7. Reproducibility and artifact publication

7.1. **Two verification tiers.** The public repository [Aut26] separates lightweight semantic checks from large proof replay.
Tier 1: source and semantics. Python 3.11.15 and the locked environment reproduce graph verification, unit tests, exact-seven frozen-state checks, and all exact-six formula reconstructions. From the repository root, the intended entry points are

```
bash scripts/bootstrap.sh
.venv/bin/python -m unittest discover \
  -s routes/finite -t . -p 'test_*.py' -v
.venv/bin/python routes/finite/check_r3_18_budget6.py
.venv/bin/python routes/finite/check_r3_18_budget6_branch1.py
.venv/bin/python routes/finite/check_r3_18_budget6_branch0_union.py
.venv/bin/python routes/finite/check_r3_18_budget7.py
```

The proof-critical Python packages are pinned to `python-flint==0.9.0`, `mpmath==1.4.1`, `python-sat==1.9.dev13`, and `six==1.17.0`. The finite route uses PySAT; the other entries are repository-wide proof dependencies.
Tier 2: large proof replay. After downloading the release assets into one directory, the content manifest is checked before decompression:

```
.venv/bin/python scripts/verify_artifacts.py \
  --directory /path/to/release-assets
```

Each matching DIMACS/DRAT pair can then be decompressed into a fresh temporary directory and replayed with the pinned `drat-trim`. The semantic checkers accept a `-drat-trim` path and replay timeout; representative commands are listed in Appendix B. A proof is accepted only when the checker exits zero and prints `s VERIFIED`.

7.2. **Why large proofs are release assets.** The six exact-six compressed files occupy about 1.26 GB, while the three raw DRAT traces occupy about 6.39 GB. Adding the smaller budget-five proofs to ordinary Git history would still make cloning needlessly expensive. The repository therefore tracks code, matrices, cut banks, semantic records, tests, the paper, and a content-addressed manifest. CNF and DRAT payloads are published as immutable release or archival assets (for example, a GitHub Release mirrored to Zenodo), keyed by SHA-256 and byte count. This design keeps ordinary clones reviewable while preserving bit-for-bit replay.

The release is complete only if it includes both the three budget-five proof pairs used by Proposition 3.1 and the three exact-six proof pairs used by Proposition 3.3. Their identities are listed in Tables 4 and 5. Incomplete, corrupt, and exploratory traces are explicitly excluded.

7.3. **Reproducibility contract.** A reproduction should report four logically distinct outcomes:

(1) the seed and artifact hashes match;
(2) semantic reconstruction passes;
(3) each required DRAT replay is verified; and

(4) the exact-seven endpoint remains UNKNOWN, unless a new, separately reviewed certificate supersedes it.

Passing only a solver run, only a hash check, or only unit tests does not by itself reproduce the full theorem.

8. Scope, limitations, and conclusion

We established a complete local statement: every triangle-free graph with independence number below 18 on the labelled vertex set of H deletes at least seven of H's 827 edges, even if additions among all original nonedges are free. The proof is carried by six checked SAT refutations: three for budget at most five and three for the exact-six layer.

The conclusion does not transfer automatically to an isomorphism class, to a different extension neighbourhood, or to a symmetric edit metric. It does not rule out a remote 100-vertex construction, and it does not improve the global bound on $R(3, 18)$. The exact-seven layer remains open. These are mathematical scope restrictions, not merely caveats about running time.

The main methodological lesson is that a computational near miss can support a rigorous negative result without being oversold as a Ramsey-number improvement. A content-addressed seed, a proof of case coverage, independently reconstructed CNF semantics, and replayable UNSAT proofs together establish a precise local barrier. Keeping bounded UNKNOWN runs outside the theorem makes the remaining target equally precise: either exhibit a seven-deletion repair and verify it directly, or produce proof-carrying refutations of all exact-seven branches.

Appendix A. Content-addressed artifact ledger

All hexadecimal values below are SHA-256 digests of the compressed files. Byte counts refer to the same compressed files and therefore permit a check before decompression.

Table 4: Budget-five proof pairs.

branch	artifact	bytes	SHA-256
0	CNF	837880	`d64f009f840597e266ccfb3b31c88f7e87eb760` `d1fc551b7ee08a0854749ba66`
0	DRAT	13125926	`60687241833df8ae957b3b4fcb02e6e9f7595c6` `e3dcbee8b77f02dad5443f913`
1	CNF	1797965	`d6d321a4ac1aeec5e99c48843a8cda9395271e8` `9555cea328e29bcda4e70ef15`
1	DRAT	37102450	`33a9e037e19beeab8dd2af4dca5ce9010d4906b` `39dc3e5cc8fd62823f518497a`
2	CNF	1363549	`8d8a543bb3cc17014941fee432673a1dc2b99c8` `cb4885d4d91bba9005f3b10a7`

branch	artifact	bytes	SHA-256
2	DRAT	21678734	`159865da9ad23eea298ebd4e9406f2f37ecc6015a593975e9d63e3b08a859897`

The corresponding uncompressed identities, independently recomputed during review, are as follows:

Branch 0 CNF: `c907dc96c1ff3378d345ead0ea22456eccef50f321cc53b6cf7dda46784d6389`

Branch 0 DRAT: `517d076efd026eab6eeb89583c0226e713be968120344e6b82c3471e0efe6b5d`

Branch 1 CNF: `5ba541f58ea42d7fb73ff51a962e98153eece0d4a8bf13c1bea3a86ed0d50429`

Branch 1 DRAT: `84a2236b7692207f59c3d2dcb3521b2f2b947428470384ac643c34a7bfbf69`

Branch 2 CNF: `441b70e023f4a9fa1837f16b1502ef3b54a2f00b6143fd5e7de0131ac95beebf`

Branch 2 DRAT: `4ff3a43bda7774c09695417a9f9684e8a856fb46bf62225a08e746fbf37d1ad6`

Table 5: Exact-six proof pairs.

branch	artifact	bytes	SHA-256
0	CNF	17600735	`47052de808b98598f2f144251a6ad73c1415dd00fef9e04b99ffe2f176229fbf`
0	DRAT	656144450	`4d948ab34f1d2475af6efe943e0a4899827c78b96c2bc3835b5161f189c26bb5`
1	CNF	5052085	`a438c9fef4d99ae516829e4405da8c9f079398501361f9c5f69f46a494ffad14`
1	DRAT	433552903	`cfe694fd728903ee9ac9f08a66ade64faf7e68763246d1d52ca29a85727494ec`
2	CNF	1205208	`c51fed43eeca3becf64bcdfbe747a4fdf3e516ce0e6a02d7d867941b87a99c06`
2	DRAT	152784369	`c9648a4e38c17dcdfe97d873095eda002abc689b3b3bd93f02fc8a369198c4b9`

For the exact-six formulas, the frozen cut-bank identities are

Branch 0, 251771 masks: `91b5709248ff641a315f5a0389b4f3fde3d38514f3b1a8b31b6cad31224f250b`

Branch 1, 63943 masks: `f8064f0f226659b4750382f781ba9a862bcc19eb49006eb8cdd2afc002ed227c`

Branch 2, 5422 masks: `aaa79a2f25d4c0d61c4467f1d0a721fd670a624e0391d96b2122723f97ba4598`

The exact-six proof records have SHA-256 identities

Branch 0: `20729b2a818aaca35cdee6e9deeb5a0e6e4bbd6dc5ca11e5abb6dfe9409b035f`

Branch 1: `4a3e7b4c729f696ff304c5f71621e018786b552146784a2141529a4c6f028666`

Branch 2: `ee0c73b80407a4dd48ca69bd1a666f391bfd5729b9a56e062deef2a6f62018de`

The source files use the following release-asset names:

```
r3_18_budget5_branch_0_twostage.{cnf,drat}.gz
r3_18_budget5_branch_1.{cnf,drat}.gz
r3_18_budget5_branch_2.{cnf,drat}.gz
r3_18_budget6_branch_0_universal_union.{cnf,drat}.gz
r3_18_budget6_branch_1.{cnf,drat}.gz
r3_18_budget6_branch_2.{cnf,drat}.gz
```

Appendix B. Independent verification protocol

This appendix states a reviewer-oriented protocol. It is intentionally more specific than the convenience wrapper in the repository.

B.1. **Seed verification.** Parse the 100×100 matrix, reject any non-binary, asymmetric, or non-zero-diagonal input, and verify its SHA-256. Count undirected edges by the strict upper triangle. Enumerate every vertex triple and record all triangles. Run the exact independent-set verifier at target 18. Repeat the structural checks for the 99-vertex principal submatrix and compare it byte-for-byte with its separately stored certificate. A successful outcome must report 827 edges and exactly the triangle $\{97, 98, 99\}$ for H, and 809 edges, no triangle, and no independent 18-set for G_0.

B.2. **Formula reconstruction.** For each exact-six branch i:

1. Assign primary identifiers $1, \dots, 4950$ to pairs in lexicographic order.
2. Generate the 161700 triangle clauses in lexicographic triple order and compare them to the DIMACS prefix.
3. Form the 826 residual deletion literals in stored seed-edge order. Regenerate the PySAT sequential equality counter at bound five. Compare all 16420 clauses and the final variable identifier 13160.
4. Compare the fixed negative unit for f_i.
5. For each remaining clause, require 153 distinct, positive primary literals. Invert the pair map, recover exactly 18 vertices, and verify that the clause is the full $\binom{18}{2}$ pair family of those vertices.
6. Compare the recovered ordered masks with the frozen cut bank and check that no unparsed clause remains.

The analogous budget-five reconstruction replaces equality at five by an at-most-four counter. It has 8238 total variables and 7394 counter clauses in each branch. The three hitting-bank sizes are 287, 15872, and 8704.

B.3. **DRAT replay.** Build `drat-trim` from source commit `2e3b2dc0ecf938addbd779d42877b6ed69d9a985`. Verify that the source tree is clean. For each of the twelve compressed assets in Tables 4 and 5, check compressed byte count and SHA-256, test gzip integrity, and decompress into a new temporary directory. Then invoke

```
/path/to/drat-trim formula.cnf proof.drat
```

and require both exit status zero and the terminal line `s VERIFIED`. Delete the temporary raw files only after recording the uncompressed hashes.

For the exact-six package, the repository checkers offer integrated replay:

```
.venv/bin/python routes/finite/check_r3_18_budget6.py \
  --drat-trim /path/to/drat-trim --drat-seconds 300
.venv/bin/python routes/finite/check_r3_18_budget6_branch1.py \
  --drat-trim /path/to/drat-trim --drat-seconds 300
.venv/bin/python routes/finite/check_r3_18_budget6_branch0_union.py \
  --drat-trim /path/to/drat-trim --drat-seconds 360
```

Time limits may need to be increased on slower hardware. A timeout is UNKNOWN, never a failed or successful proof.

B.4. **Logical audit checklist.** A reviewer can check the mathematical implication independently of the code:

1. the three f_i are precisely the edges of a seed triangle;
2. every triangle-free final graph omits at least one f_i;
3. original nonedges are primary variables but not deletion literals;
4. every installed C_S is entailed by $\alpha(F) < 18$;
5. omitting uninstalled C_S clauses makes the CNF weaker, so an UNSAT result remains sufficient;
6. the budget-five refutations force any putative budget-six candidate to have exactly six deletions;
7. after choosing one absent f_i, exactly five residual deletion literals are true; and
8. checked UNSAT of all three exact-six branches contradicts the model provided by any candidate graph.

Appendix C. Frozen exact-seven first-round record

All three endpoints are UNKNOWN. The branch records are content-addressed as follows.

Table 6: Exact-seven bounded-search records.

branch	result JSON SHA-256	checkpoint SHA-256
0	`97e777da32b82a83b8e999c9af1f92 7c87b67689b8fb72f8b31ded07b39a 2f7f`	`918dc1cc80860d97b3209f6d969a15 6b9df5fb8b4df7f0fa910e72fdc577 8098`
1	`81734a23298ea9c3364a047c3b7b10 910bcd44faf9ff55c3a251fd709f6a c64f`	`8491cbd5e073170b6f7a503efb3d6a f6c077b6d72defbf9cd6feb6f06e85 fc7f`

branch	result JSON SHA-256	checkpoint SHA-256
2	92e2c948892d6f298e0ca3a5055cf3 1cf3eea361b6eee8197ba536922e43 7554	9e8ea23904378ec1bfc6e4c486fd32 f10ccc99c1646855c7589c21cc3c9b 9f5c

The frozen-state checker, its unit tests, and the branch generator have respective SHA-256 identities

checker: `8c1f2629193de8b16e018f036ce462b8e18110b9f4f1c13b3162b25fe415d6f4`
tests: `46b0324d67baaf66bae49027bf4ddbfe3aa08684d3cbe364326bc4400914c987`
generator: `13a7cfeb564297fcad2ea2c6ca831a6bf4b4755e0223abcf9d1bd1178103d0de`

Each branch loaded the common universal bank in under six seconds and then spent approximately 300 seconds in its first solver call. The registered endpoints were

branch	fixed absent	solver	endpoint
0	$(97, 98)$	CaDiCaL	UNKNOWN
1	$(97, 99)$	Glucose 4.2	UNKNOWN
2	$(98, 99)$	MapleChrono	UNKNOWN

The state checker verifies formula dimensions, the universal bank digest, the exact counter truth table, exclusion of original nonedges from the counter, and the three timeout records. It does not transform those timeout records into a satisfiability or unsatisfiability claim.

References

[Aut26] Anonymous Author(s). Reproducible ramsey bounds: source, certificates, and artifact manifests. Software and data repository, 2026. Archived release identifier to be added at publication.

[BFF+24] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, Nils Froleyks, and Florian Pollitt. CaDiCaL 2.0. In *Computer Aided Verification—CAV 2024*, volume 14681 of *Lecture Notes in Computer Science*, pages 133–152. Springer, 2024.

[Heu16] Marijn J. H. Heule. The DRAT format and DRAT-trim checker. *arXiv preprint*, 2016.

[IMaMS18] Alexey Ignatiev, António Morgado, and Jo ao Marques-Silva. PySAT: A Python toolkit for prototyping with SAT oracles. In *Theory and Applications of Satisfiability Testing—SAT 2018*, volume 10929 of *Lecture Notes in Computer Science*, pages 428–437. Springer, 2018.

[Rad26] Stanisław P. Radziszowski. Small Ramsey numbers. *The Electronic Journal of Combinatorics, Dynamic Survey DS1*, 2026. Revision 18, 24 April 2026.

[Sin05] Carsten Sinz. Towards an optimal CNF encoding of Boolean cardinality constraints. In *Principles and Practice of Constraint Programming—CP 2005*, volume 3709 of *Lecture Notes in Computer Science*, pages 827–831. Springer, 2005.

[WHH14] Nathan Wetzler, Marijn J. H. Heule, and Warren A. Hunt, Jr. DRAT-trim: Efficient checking and trimming using expressive clausal proofs. In *Theory and*

Applications of Satisfiability Testing—SAT 2014, volume 8561 of *Lecture Notes in Computer Science*, pages 422–429. Springer, 2014.